



THE PRODUCT OF (CAUSAL) QUALITATIVE CODING CAN BE A MODEL YOU CAN QUERY

The result of qualitative coding of texts: is it a model?

Every type of Qualitative Data Analysis (QDA) is a bit different....

1. The most important result of coding might be a **deeper understanding of the text**. Something the researchers have in their heads. Perhaps something shared within a group. They can answer new questions about the data and write summaries of it from a new angle. In a sense you could say they have more or less formally developed a **model** of the data or even a theory about it, or, in the sense of Grounded Theory, a theory which goes beyond that data but is partly inspired by it. If you yourself read some of these different outputs and engage with them, maybe you yourself can start to build such a model in your head.
2. Apart from the intangible products in researchers' heads there are also **tangible products such as research reports**. Different schools of QDA give different importance to tangible as opposed to intangible outputs.
3. Thirdly, there may be **tabular outputs** such as tables of coding frequencies, cross-tabulations, etc. The definitive set of tables can be queried to answer additional questions like "how many women who mentioned working from home also see television as a an outdated medium". These kinds of tabular outputs can be quite useful to answer different questions.

So when you do an evaluation, what's the product? What do you get?

Obviously, you can think of the evaluation report, which might answer predetermined questions, but it may also include material that goes beyond the specific questions we were tasked with answering—for example, to address unanticipated issues or simply to describe or contextualize. Another important output is relational: hopefully, people have come together in a way that helps to expand learning and perhaps develop projects or relationships.

But today I want to talk about something different.

A statistical analysis not only answers questions but gives you the whole model — can a qual analysis do that too?

When you do quantitative research, you might have specific research questions, but often one of the major outputs is a statistical model of the phenomenon. (There might be an effort to go beyond the data

and hope that the model generalizes more widely, but that's not my focus here.) In the simplest case, the model might represent a suspected causal relationship—say, between the amount of screen time in the evening and difficulty falling asleep.

At the very least, the model allows us to look at a case in the dataset and say: on this day, this person looked at a screen for three hours and rated, let's say, a difficulty of four out of five falling asleep on some self-rating scale. Because we have the model, **we can explain that**: yes, this is quite a high level of difficulty, and it's explained at least partly by a high level of screen time, at least in this individual case. The model might also enable us to **make predictions**, like: people who, at least in this context, spend more than three hours on screens in the evening are, on average, going to experience a higher level of difficulty falling asleep.

A more sophisticated model will probably capture more variables, and many models—like directed acyclic graphs—link up these kinds of connections into a causal network, so you can explain or even predict how tweaking one variable will affect another variable downstream of it. Of course, there are other kinds of statistical models apart from causal models, but if you're reading this, you love causal models, don't you?

There's a relatively small but extremely well-funded section of evaluation activity based around this kind of statistical causal model, with randomized control trials (RCTs) as one facet. Ideally, an RCT is tasked not primarily with producing a model, but with answering a specific question, like: which is the better of these two interventions? Or: does this treatment work better than a placebo? But these are calculations conducted on the underlying model, which, from the point of view of workflow, is the major output of the work.

To generalise or not to generalise

There are two ways you might want to use that kind of model.

- One is to answer further questions about the same dataset—for example, to ask whether a particular subset (say, people over 70) differ in how screen time influences difficulty falling asleep, compared to other subgroups.
- If it's a sophisticated model, it might allow us to *generalize beyond* this specific context, perhaps by including more general variables like attention style or eye movement speed, which might help explain and predict behavior in other contexts.

Most quantitative researchers make a big deal about generalizing the model beyond the specific use case or context. In fact, the whole point of the study is normally to do that, and there's a whole armoury of tools, concepts, and arguments about how and under what conditions you can generalize a model to other people, other years, or even other countries with different kinds of screens, etc.

Can qualitative research do that?

The majority of evaluations aren't like that, although they might include a specific quantitative question somewhere in their terms of reference. In most cases, we think of the research output as a report in which the original (possibly modified) list of questions is answered, with additional narrative to summarize and link these sections.

Now, going beyond strictly quantitative paradigms, some evaluation projects will also include what we might call **qualitative modelling**. For example, if we're using QCA (Qualitative Comparative Analysis), apart from answering specific questions, we've likely also produced QCA-style tables, which could help us answer other questions beyond those we were actually tasked with. You might see those tables as annexes to the report. The same goes for causal loop diagrams and other techniques, which are essentially quantitative models but with a more restricted set of numbers. For example, in causal loop diagrams, we might model a variable like inflation with a number from -1 through 0 to +1, and do the same for variables like unemployment or military threats, building models of the relationships between these things using simplified numbers.

What I want to argue here is:

any halfway decent evaluation, which at least implicitly gathers qualitative information about how things within the evaluation influence one another, can be considered as constructing a qualitative causal model. This is irrespective of the specific methods used—even if it doesn't include something explicitly called causal pathways analysis or causal mapping.

Qualitative models as products: theory, model, or something else?

In quantitative research, the idea of a "model" as a product is well established: you build a model, and then you can query it to answer new questions—even ones you didn't anticipate at the start. But what about qualitative research? Can the result of a qualitative analysis be a model in this sense, rather than just a set of answers to specific questions or a summary?

Of course (some) qualitative research produces models. Just don't call them that.

Some qualitative researchers do indeed conceptualize their results as models. For example, grounded theory often produces a theoretical model that explains the underlying processes or relationships within the data. These models can be revisited and "queried" to generate new insights beyond the initial research questions.

However, many qualitative researchers are more comfortable with the term "theory" rather than "model." "Theory" aligns more closely with the interpretive and conceptual nature of qualitative work, emphasizing explanation and understanding rather than prediction or parameterization. Still, the distinction is often

more about language than substance: both models and theories can serve as frameworks for making sense of data and for generating new questions.

How are qualitative models used?

In qualitative research, especially in fields like grounded theory and narrative inquiry, the focus is less on "prediction" and more on generating understanding or insight. Researchers talk about "theorizing" from the data—developing concepts and frameworks that explain the phenomena under study. Once a theory or model is developed, it can be revisited, interrogated, and applied to new data or different contexts. This iterative process allows for continual refinement and deeper insight.

Importantly, a qualitative model or theory can also be used to answer new questions about the same dataset. For example, after developing a grounded theory, researchers (or others) can return to the theory and use it as a lens to interpret further cases, refine concepts, or generate new insights—without having to re-examine all the original data. This practice is sometimes referred to as "secondary analysis" or "theoretical application," where the theory or model functions as a standalone analytical tool.

In causal mapping, for instance, the model might consist of a network of causal links derived from qualitative data. Even if there are no quantitative parameters on the links, the model can still be queried: "Is there evidence for a causal pathway from A to B?" or "What are the main factors influencing outcome X?" This allows the model to be used flexibly, supporting both anticipated and unanticipated lines of inquiry.

Why does this matter?

Thinking of qualitative research outputs as models (or theories) that can be queried and reused has several advantages:

- **Transparency:** It makes explicit the structure of the findings and how they relate to the data.
- **Reusability:** Others can use the model/theory to answer new questions or apply it in new contexts.
- **Iterative learning:** The model can be refined and expanded as new data or perspectives emerge.
- **Bridging paradigms:** It helps bridge the gap between qualitative and quantitative traditions, showing that both can produce structured, interrogable outputs.

In summary, while qualitative researchers may prefer the language of "theory" over "model," the idea is the same: a well-constructed qualitative analysis can produce a framework that is more than just a set of answers—it is a model of the phenomenon, one that can be queried, shared, and built upon.